

22) $\int_{-1}^1 (1 + \sqrt{1-x^2}) dx$

Sol:

$$y = 1 + \sqrt{1-x^2}$$

$$y-1 = \sqrt{1-x^2}$$

$$(y-1)^2 = 1-x^2$$

$$x^2 + (y-1)^2 = 1$$

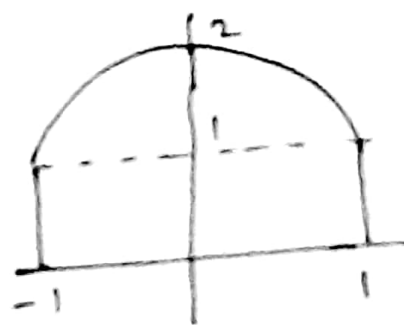
Centre = (0, 1)

Radius = 1

Area of Semi-circle = $A_1 = \frac{\pi r^2}{2} = \frac{\pi(1)^2}{2} = \frac{\pi}{2}$
(Upper part)

Area of rectangular base = $A_2 = l \times w = (2)(1) = 2$

Total Area = $A_1 + A_2 = \frac{\pi}{2} + 2 = \boxed{\frac{4+\pi}{2}}$ ^{Square units}
Ans



28) ~~Use~~ Evaluate the definite integral of $f(x) = 3x + \sqrt{1-x^2}$ using known area formula.

on (b) $[-1, 1]$.

Sol: $\int_{-1}^1 (3x + \sqrt{1-x^2}) dx = \int_{-1}^1 3x dx + \int_{-1}^1 \sqrt{1-x^2} dx$
 $= \int_{-1}^0 3x dx + \int_0^1 3x dx + \int_{-1}^1 \sqrt{1-u^2} dx$

$= \int_{-1}^0 3x dx + \int_0^1 3x dx + \int_{-1}^1 \sqrt{1-u^2} dx$

 $= \frac{1}{2}(1)(-3) + \frac{1}{2}(1)(3) + \frac{\pi(1)^2}{2} = \boxed{\frac{\pi}{2}}$
 Ans

